

M. AMINE ATOUI, PH.D.

Bayesian & AI-Driven Decision Support · Quality, Reliability & Automation Engineering

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Profile

Researcher and lecturer working at the junction of quality and reliability engineering and artificial intelligence. My Ph.D. thesis combined statistical quality control with Bayesian networks for industrial fault detection and diagnosis, and that combination, data-based methods paired with domain knowledge, has stayed the throughline of my work since: decision-support methods spanning classical statistical process control through to modern Bayesian, causal and deep-learning approaches, applied to real industrial, energy and embedded systems. I have taught and coordinated courses across that same range, from statistical quality control, metrology and automation to postgraduate Bayesian statistics for machine learning (93% student evaluation, a 30-point improvement year-on-year). Over 15 years across CNRS, Polytech, the University of Lille, Institut Mines-Télécom, Aix-Marseille University, the University of Oulu, the University of South Brittany and Halmstad University, I have secured or contributed to more than €3.4 million in competitive research funding, including €2.8 million as Principal Investigator or Co-Principal Investigator, and published 60+ peer-reviewed works across journals, conferences and technical reports.

At a Glance

- €3.4M+ in total competitive research funding since 2018, including €2.8M as PI or Co-PI (€100k as sole PI, Green Bridge Island; €2.7M as Co-PI, Solbruk)
- 60+ peer-reviewed publications across journals, conferences and technical reports
- 950+ hours of university teaching across 6 institutions in 3 countries, spanning statistical quality control, automation, and Bayesian statistics for machine learning
- 3 Ph.D. students co-supervised (1 defended, 2 in progress); 3 industrial Master's theses supervised; 40+ engineering internships supervised across Europe
- Interview panel member for two Ph.D. student recruitments, Halmstad University
- Research collaborations spanning 9 countries: France, Sweden, Finland, Portugal, Italy, Poland, China, USA, Canada
- Qualified by the French National Scientific Committee (CNU) for Associate Professor positions, Section 26 (Applied Mathematics) and Section 61 (Computer Engineering, Control & Signal Processing)

Education

Ph.D. in Engineering Sciences (spécialité Productique)

2011 – 2015

LARIS, Polytech Angers, University of Angers, France

Thesis: "Contributions to Fault Detection and Diagnosis using Bayesian Networks." Hybrid probabilistic framework combining data-based (statistical quality control) and knowledge/physics-based (automation, control theory) methods for industrial fault detection and diagnosis, describing and explaining the state of an installation at a given time, whether normal operation or a fault state due to sensor, actuator or system degradation. Qualified for Associate Professor positions in CNU Section 26 (Applied Mathematics) and Section 61 (Computer Engineering, Control and Signal Processing).

M.Sc. in Pattern Recognition & Optimization

2009 – 2011

University of Annaba, Algeria, ranked 1st of 162 students

Core: optimization, signal processing, game theory. Applications: industrial, medical, computer vision, biometrics.

B.Sc. in Computer Science (Applied Mathematics track)

2006 – 2009

University of Annaba, Algeria, and Université Côte d'Azur, Nice, France

Mathematical analysis, algebra, distributed networks, programming and algorithmics, computer architecture, complex systems, automata.

Academic & Research Positions

Independent Consultant, AI for Product & Asset Decisions

2024 – present

Self-employed, remote

Advising industrial and energy clients on AI-driven decision support and predictive maintenance. Maintained an active research output through this period: co-supervision of Guangqiang Li's Ph.D. at Wuhan University, China (extended to 2026), and five peer-reviewed papers published in 2024–2025 on open-set and domain-generalizing fault diagnosis.

Researcher, Applied-AI Engineering

2022 – 2024

ITE Department, University of Halmstad, Sweden

Contributed to CAISR+ (open-set and domain adaptation, meta- and multi-task learning, multi-modal learning, survival analysis, synthetic data) and mentored a postdoctoral researcher on iHELM (explainable AI for the maritime sector), extending earlier XPM work. Principal Investigator on Green Bridge Island (2023–2024) and Co-Principal Investigator on Solbruk (2024–). Coordinated and taught the Bayesian Statistics for Machine Learning course, restructured from the ground up to a 93% evaluation (a 30-point improvement on 2022), and led restructuring initiatives within the IT department. Actively participated in the SAFER and REBEL research programmes and contributed to proposal writing for ProActS, applying for funding from VR, VINNOVA and INTERREG Europe as both PI and co-PI. Supervised Master's students and co-supervised Ph.D. students; sat on interview panels for postdoctoral, Ph.D. and research engineer recruitment.

Researcher, XPM (eXplainable Prescriptive & Predictive Maintenance)

2022

IMT Nord Europe, France

Designed an explainable AI decision-support framework generating maintenance recommendations from failure modes, degradation severity, dynamic evolution and spare-part inventory, with trigger parameters tied to environmental and operational conditions, component reliability and criticality, and Remaining Useful Life, aimed at improving operators' understanding of asset health. Coordinated and participated in cross-partner workshops. Case studies in EVs, subway trains, steel mills and wind farms, with the University of Porto, Jagiellonian University and Halmstad University.

Researcher, Hybrid Diagnostics for Embedded Networks

2020 – 2021

CNRS IEMN (TELICE) & CNRS CRISAL (ToSyMa), University of Lille, Centrale Lille, Université Polytechnique Hauts-de-France, France

Recruited to bridge two research laboratories and develop new online hybrid fault-detection approaches, combining available information, data, models and knowledge, for embedded networks in transport vehicles, co-financed by the ERDF, the French government and the Hauts-de-France Regional Council. Multidisciplinary work spanning electronics, transmission, communication, control and diagnostics. In parallel, continued theoretical development of statistical, causal and deep-learning methods for robust, explainable decision-making. During the COVID-19 period, produced 10+ papers as first or second author and taught approximately 100 hours (Nov. 2020 – Apr. 2021).

Senior Lecturer, Lab-STICC CNRS

2018 – 2020

University of South Brittany, France

Contributed to PRECOGS (acquiring and analysing fighter-jet pilot workload, stress and fatigue) and HUMANISM (human-cobot dialogue rules for shared box-handling tasks), and explored digital twins for machine reconfiguration, preventive maintenance and cybersecurity-aware anomaly detection. Close to 500 hours per year of teaching and supervision at the Bachelor of Technology in Quality, Logistics and Industrial Organization.

Postdoctoral Fellow, Prognostics & Health Management

2018

LIS CNRS, Polytech Marseille, Aix-Marseille University, France

Contributed to FUI-19 MMCD, developing hybrid methods and algorithms for life-cycle analysis and residual-life estimation in aircraft and vehicle embedded systems. Proposed a novel probabilistic approach identifying the influence of condition-monitoring variables on dynamic strain aging; Remaining Useful Life estimated via temporal projection of health-index dynamics to a predefined failure threshold, accounting for future operating modes and current degradation.

Postdoctoral Fellow, Decision-Making for Product Development

2015 – 2017

M3S Group, University of Oulu, Finland

Contributed to the FiDiPro/VALUE programme (Finland Distinguished Professor Programme) improving decision-making for software-intensive products and services with Nokia, Ericsson, F-Secure and Bittium: built a Python/Django tool for partners to weigh the impact of features and requirements on value factors, identified the values partners use intrinsically in decision-making, and helped design a Bayesian framework aggregating decision-making data into an overall value estimate for a proposed action. Deepened expertise in Bayesian methods for prediction and decision-making combining expert knowledge and collected data.

Assistant Professor (ATER) & Ph.D. Candidate

2011 – 2015

Doctoral research on hybrid fault detection and diagnosis, combining data-based (statistical quality control) and model-based (automation and control theory) methods; required a theoretical understanding of model-based, data-based, hybrid, fusion and Bayesian methods. Teaching assistant and, from 2014, Assistant Professor in quality, innovation and reliability engineering, preparing course materials and delivering lectures, discussions and assessments across polytechnic subjects and levels.

Research

My work combines classical quality and reliability engineering with Bayesian, causal and deep-learning methods to build decision-support systems whose outputs can be explained, trusted and acted on, whether that action happens on a factory floor, a maintenance schedule, or a sensor/actuator/control loop. I am extending this into multi-modal, federated and robotics/mechatronic settings, while continuing to apply the same hybrid, data-plus-domain-knowledge approach to statistical process control and reliability problems more broadly.

Selected Grants & Funding

Solbruk

Co-Principal Investigator · 2024 – 2027

Cross-border Interreg Öresund-Kattegat-Skagerrak Living Lab combining agrivoltaics (solar cells, crops, agricultural machinery and storage) across four demonstration sites in Sweden and Denmark, led by Hushållningssällskapet Halland (Principal Investigator) with RISE, Region Halland and European Energy A/S among the partners. I represent the ITE department as Co-Principal Investigator, alongside Pia Ulvenblad (Business, Innovation & Sustainability dept., Halmstad University). My role: AI-driven control of solar-farm operations, real-time parameter adjustment from weather, satellite and soil-sensor data, predictive maintenance of equipment, identification of the most profitable crops for specific environments, and AI-enhanced autonomous drones for efficient navigation within solar-farm parks. Total budget €2,703,457; EU funding €1,622,056. [Official project listing](#).

Green Bridge Island

Principal Investigator · 2023 – 2024

Interreg Öresund-Kattegat-Skagerrak pre-project (ref. I20364015) with Øresundsbro Konsortiet, evaluating a 6.5 MW battery installation at the Öresund Bridge alongside its 2.5 GWh of annual solar production, to strengthen cross-border energy robustness between Denmark and Sweden. As sole Principal Investigator, I designed the AI approach for optimising when the storage is charged, discharged or held in reserve, using predictive models and risk-aware cost-benefit analysis of competing strategies, and identified the cross-border partners needed for a full-scale follow-on. Total budget €100,000; accepted with a follow-on commissioned contract.

XPM, CHIST-ERA

Researcher · 2022

Explainable AI decision-support for predictive and prescriptive maintenance, with IMT Nord Europe, University of Porto, Jagiellonian University and Halmstad University. Not a PI role; contributed as researcher responsible for framework design. Consortium funding €604,236 over 24 months.

CAISR+ & FREEPORT

Researcher · 2022 – 2024

Multi-task and multi-modal learning, survival analysis and synthetic data generation for predictive maintenance, with Volvo Trucks. FREEPORT (Federated Learning and Edge Processing for Safe and Efficient Operation) applies federated learning and edge processing to fleet operation. Not a PI role. Funded by KK-stiftelsen (the Knowledge Foundation).

iHELM

Researcher & mentor · 2022 – 2024

Explainable AI for the maritime sector, with Volvo and CetaSol AB; mentored a postdoctoral researcher on the project. Not a PI role. Funded by VINNOVA.

ELSAT2020 / DIACA

Researcher · 2020 – 2021

Hybrid (information, data, models, knowledge) fault detection and localisation for embedded transport networks. Not a PI role; recruited as researcher to bridge CNRS IEMN and CRISTAL. Funded by the ERDF, the French State and the Hauts-de-France Region.

FUI-19 MMCD

Researcher · 2018 (funded 2015)

Next-generation avionics cockpit display screens, with industrial partners IRTS and NOLAM and research labs INRIA and LSIS. Not a PI role; my contribution was hybrid life-cycle and Remaining Useful Life estimation for the embedded aircraft/vehicle systems involved. Funded by the French State and regional/territorial authorities (specific amount not disclosed to project researchers).

VALUE, FiDiPro

Researcher · 2015 – 2017

Decision-making tools for software-intensive product development, with Nokia, Ericsson, F-Secure and Bittium. Not a PI role. Funded by TEKES / FiDiPro Programme, University of Oulu.

Also: proposal work as PI or co-PI submitted to VR, VINNOVA and INTERREG Europe beyond the funded projects above, including a national proposal on embedded wired-network monitoring currently under revision for resubmission.

Selected Publications

- G. Li, M. Amine Atoui, X. Li (2025). Attention-based multi-scale temporal fusion network for uncertain-mode fault diagnosis in multimode processes. *Process Safety and Environmental Protection*.
- G. Li, M. Amine Atoui, X. Li (2025). Dual adversarial and contrastive network for single-source domain generalization in fault diagnosis. *Advanced Engineering Informatics*.
- G. Li, M. Amine Atoui, X. Li (2025). Open-set fault diagnosis in multimode processes via fine-grained deep feature representation. *arXiv:2508.09463*.
- G. Li, M. Amine Atoui, X. Li (2025). Fault diagnosis across heterogeneous domains via self-adaptive temporal-spatial attention and sample generation. *Knowledge-Based Systems*.
- C. Lou, M. Amine Atoui, X. Zhang, D. Xu, H. Zhong (2025). Graph neural network-based semi-supervised open-set fault diagnosis for marine machinery systems. *Intelligent Systems with Applications*.
- C. Lou, M. Amine Atoui, X. Li (2024). Recent deep learning models for diagnosis and health monitoring: a review of research works and future challenges. *Trans. Inst. Meas. Control*.
- M. Abuella, M. Amine Atoui, S. Nowaczyk, S. Johansson, E. Faghani (2024). Spatial clustering approach for vessel path identification. *IEEE Access*.
- M. Amine Atoui, V. Cocquempot (2023). Explainable root cause and pathway analysis with robust and adaptive statistics. *Computers in Industry*.
- M. Amine Atoui, A. Cohen, V. Cocquempot (2021). Unlocked decision making based on causal connections strength. *European Journal of Control*. Recommended for publication by Prof. Thomas Parisini.
- M. Amine Atoui, A. Cohen, S. Verron, A. Kobi (2019). A single Bayesian network classifier for monitoring with unknown classes. *Engineering Applications of Artificial Intelligence*.

Full list of 60+ publications on Google Scholar and available on request.

Teaching

I design, deliver and evaluate courses with a strong applied focus, moving students from theory to hands-on labs and prototypes, and have actively contributed to curriculum development across every institution I have taught at. My Bayesian Statistics for Machine Learning course at Halmstad University was restructured from the ground up and received a 93% evaluation in 2023, a 30-point improvement on the prior year.

Applied Bayesian Statistics & Probabilistic ML

Halmstad University · 2023

Commissioned course (approx. 30h) for Master's and Ph.D. students and industry professionals, several from RISE and the University of Amsterdam. Fully online, with weekly lectures and Python/Jupyter labs over 7 weeks: Bayesian concepts, priors, inference, parametric models, sampling-based methods, sequential inference (Kalman/particle filters), variational inference, model selection and Bayesian deep neural networks. Built a dedicated course space with lectures, labs and a student collaboration area; defined clear feedback timelines; grades reviewed by a colleague for accuracy. Evaluation: 93%.

Guest Lecturer, Control Theory & C Programming

University of Lille (IUT de Lille, EEA Dept.) · 2021

Approx. 100 hours delivered on request from colleagues across departments, alongside a research contract: functions, data structures, pointers and recursion in C; introductory control theory and Matlab-based system identification.

Senior Lecturer, Quality, Logistics & Industrial Organization

2018 – 2020

IUT QLIO, University of South Brittany, France

Approx. 450–500 hours as responsible coordinator and examiner: statistical quality control, metrology, operations research, reverse logistics, flow management, reliability and maintenance, production and multi-module student projects. Full concept-to-prototype design-build studio in the Innovation & Technological Hall. Mentored apprentices, advised juries, and participated in workshops, forums and open days.

Lecturer, Quality, Innovation, Reliability & Numerical Analysis

2012 – 2015

Polytech Angers, France

Teaching assistant (2012–2014, 2×64h/year) and lecturer (2014–2015, 200h): quality control and sampling plans, statistics, statistical quality control (Shewhart, EWMA, CUSUM), industrial economics and process optimization, and **numerical analysis** (interpolation and approximation, numerical integration, solving linear,

non-linear, ordinary differential and partial-derivative equations). Supervised student-led industrial projects.

Supervision

I supervise via regular meetings, expert feedback on research design, data analysis, experimental setup and results interpretation, supporting publication efforts and preparation for industry or academia. One undergraduate project I supervised, on sustainability and local products, received regional media coverage (*Ouest-France, Le Télégramme*).

Ph.D. co-supervision

- Guangqiang Li: meta-learning and domain generalization in multi-mode settings. Wuhan University, China, with Prof. Xiangshun Li. 2023–2026.
- Canberk Özen: graph neural networks in federated learning for anomaly detection. Halmstad University, with Slawomir Nowaczyk and Prayag Tiwari. 2023–2024.
- Chuyue Lou: transfer and deep learning for anomaly detection and diagnosis. Wuhan University of Technology, with Prof. Xiangshun Li and Prof. Jin Jiang. 2019–2023, defended.

Industrial Master's theses

- Omar Alfakir: estimating ambient temperature using Attention-LSTM models. SONY Lund, 2023.
- Markus Bürgschwendtner: contrastive learning for anomaly detection in industrial production. DAZ, Austria, 2023.
- Stefan Falkner: IMU integration for a prosthetic-arm control system, with Chalmers University. Center for Bionics and Pain Research, Sweden, 2024.

Plus around 40 undergraduate and postgraduate engineering-student internships supervised, visited or examined in research labs and industry across Europe.

Service & Academic Leadership

Recruitment & governance

- Interview panel member for two Ph.D. student recruitments, Halmstad University
- Preliminary evaluation and interviews for Postdoctoral Researcher, Ph.D. Student and Research Engineer positions, Halmstad University (2022–2024)
- Academic opponent for undergraduate and graduate students: feedback and critical questions on internships, theses and group projects
- Elected member, Scientific Council of LARIS (2013–2015)
- Doctoral student representative, HCERES evaluation and LARIS/LASQUO-LIS group merger (2014–2015)
- Weekly, monthly and annual reporting for projects led or contributed to; review and validation of Ph.D. Individual Study Plans; course scheduling and exam/grade reporting across institutions

Session chair & conference service

- Co-chaired sessions on fault tolerance, diagnosis, and fault-detection & tolerant control at MED, IEEE-ICCA, ECC and ICINCO: reviewing submitted slides/videos, preparing questions, introducing speakers and moderating
- Responsible for the “Decision Making” booth, N4S2016 workshop, Rovaniemi, Finland

Outreach & committees

- Organiser/participant, La Fête de la Science, Pays de la Loire, Bretagne, Hauts-de-France and Oulu (Finland)
- English Days, open days, public fairs and orientation forums; jury committees for student evaluations, examinations and graduations
- Working and brainstorming groups for new courses, curricula and specialized programmes

Reviewer, international journals

Automatica, Expert Systems with Applications, Engineering Applications of Artificial Intelligence, ISA Transactions, Journal of the Taiwan Institute of Chemical Engineers, Knowledge and Information Systems, Transactions of the Institute of Measurement and Control, Journal of Nuclear Engineering and Design, Journal of Industrial Information Integration, The Journal of Supercomputing, Statistics and Computing, Journal of Software: Evolution and Process, MDPI journals, Measurement, Scientific Reports, Sensors, Qeios, and others.

Reviewer, international conferences

IEEE International Conference on Systems, Man, and Cybernetics; IFAC World Congress; IFAC SafeProcess; European Control Conference; International Conference on Informatics in Control, Automation and Robotics; International Conference on Control, Automation, and Systems; International Conference on Modern

Management based on Big Data; and others.

Distinctions

- French National Scientific Committee (CNU) qualification, Associate Professor tracks, Sections 26 and 61
- Invited by the Dean of the Wuhan School of Automation (Dec. 2023) to co-advise Ph.D. and Master's students; separate invited research visits to Wuhan University of Technology (Prof. Xiangshun Li) and Western University, Canada (Prof. Jin Jiang)
- Multiple invitations to serve as guest editor or contribute special-issue papers at MDPI journals, declined
- Best Ph.D. Creative & Innovative Project, UNAM Doctoriales (2013)
- Ranked 1st of 162, M.Sc. Pattern Recognition & AI cohort, University of Annaba (2011)
- Mundus Scholarship, Université Côte d'Azur (2008); Ph.D. grant, Région Pays-de-la-Loire
- Invitation to the EJC special issue for best ECC 2021 papers (Prof. Thomas Parisini)

Collaboration & Industry Engagement

Universities & research institutes, 9 countries

RISE Research Institutes of Sweden (via the MAISTR course and the FREEPORT project), ProActS and the SAFER (Chalmers vehicle & traffic safety) and REBEL (smart living, invited by Prof. Vaike Fors) research programmes at/with Halmstad University, University of Torino (Italy), University of West Florida (USA; A. Cohen, Ph.D.; statistical signal processing applied to healthcare), Wuhan University of Technology (invited by the Dean of the School of Automation to co-advise Ph.D./Master's students; separate research-visit invitation from Prof. Xiangshun Li), Zhejiang University of Finance and Economics (China; meta-learning applied to predictive maintenance for multi-modal industrial systems, continuing the collaboration with Dr. Chuyue Lou), and Western University, Canada (invited research visit, Prof. Jin Jiang).

Industry, with funding context

- Sony Lund, Sweden — Alfakir's Master thesis; funded by Sony Lund
- Ericsson — FiDiPro project; funded by the Academy of Finland
- Øresundsbron — Green Bridge Island; funded by Interreg Öresund-Kattegat-Skagerrak
- Volvo Trucks — CAISR+ and FREEPORT; funded by KK-stiftelsen
- Volvo & CetaSol AB — iHELM; funded by VINNOVA
- Hushållningssällskapet — Solbruk; funded by Interreg Öresund-Kattegat-Skagerrak
- HMS Industrial Networks AB — project submission to VINNOVA
- F-Secure, Bittium, Nokia, Naval Group, Metro do Porto, ArcelorMittal — various research, education and supervision projects since 2012, funded by the Academy of Finland, the French government, French research agencies, French regional councils, EU funds and Bpifrance

Skills & Languages

- **Programming:** Python, C/C++/C#, MATLAB/Simulink, R, Java, .NET, JavaScript, Lua, Perl, UML, LaTeX
- **Machine learning:** PyTorch, TensorFlow, Keras, scikit-learn, Jupyter, NLTK; deep and probabilistic ML, causal inference, survival/RUL analysis
- **Bayesian network software:** Hugin, BayesiaLab, AgenaRisk, BNet, Analytica
- **Other tools:** Django/PostgreSQL, PLC/Ladder, GRAFCET, FSM, Statgraphics, Statistica, Moodle
- **Languages:** French (fluent), English (fluent), Finnish (basic), Swedish (elementary)

Project documentation, funding decisions and referee contacts available on request.